

DIPANWITA MANDAL

Software Engineer | Generative AI

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SUMMARY

Software Engineer with **4+ years** delivering **React.js/Next.js** products and **full-stack** features with **Node.js/Nest.js** in high-growth startups. Shipped **4 production dashboards in 1.5 weeks** and **3 emissions systems in 72 hours**, led a **0→1 EC BOQ** module with editable tables and calculators. Cut API boilerplate **40%** with **TanStack Query**, standardised large-dataset tables with **Material React Table** (**−60%** feature lead time), built a **JSON-driven form engine (10+ fields, −65%)**, and shipped **real-time ECharts** visualisations (**−70%** UI defects). Mentored juniors, drove code reviews/CI, and partnered with design for a **50%** faster design-to-prod cycle. Recent work includes Net-Zero forecasting (**Next.js + Nest.js**) and **LLM features** using **LangChain/RAG**.

WORK EXPERIENCE

Software Engineer – Fullstack

Accacia

FEB 2023 - PRESENT

(2 YEARS 8 MONTHS)

- Shipped **4 MVP dashboards in 1.5 weeks** and **3 emissions-tracking systems in 72 hours**, enabling real-time carbon reporting across **3+ asset types**.
- Built **EC BOQ (0→1)** with editable tables, live calculators, and charts; delivered to production in **2 months**.
- Introduced **TanStack Query** for API caching/sync → cut boilerplate by **40%** and improved perceived latency.
- Developed reusable **Material React Table** components for large datasets → reduced feature build time **60%** and standardized UX.
- Created a **JSON-driven form engine** supporting **10+ field types** → saving future form work by **65%**.
- Implemented a **real-time visualization layer** with **ECharts** → lowered UI defect rate **70%** via live updates.
- Defined **REST API contracts** with backend → integration issues dropped **25%** and stability increased.
- Mentored **2 junior engineers**, led code reviews and component standards → team velocity **+20%**.
- Ran **Figma → React** workflow, shortened design-to-production cycle **50%** through tighter collaboration.

Projects at Accacia

- Net Zero Target & Forecasting (Next.js + Nest.js)**: Configuration of baseline/target years, portfolio-level cascade of targets, **dynamic emissions projections**, cost-abatement models, and strategy dashboards for real-time decarbonization tracking.
- On1y Homes (React + Supabase)**: Interior design platform for project listings, client interactions, and design requests; achieved **500+ active users in month 1**.
- Studio.On1yHomes (AI Interior Analysis)**: Added AI-driven recommendations for space planning, automated design reports → **turnaround time −50%**.

Senior Software Development Engineer

KAFQA

Jun 2022 - Dec 2022

(7 MONTHS)

- Integrated **Mixpanel** with **100% coverage** of critical click events → enabled data-driven roadmap decisions.
- Established the foundation of a **CRM platform** with **15+ reusable React components** → reducing future dev time by **30%**.
- Optimized frontend performance via refactors and state management → **page load −15%**.

Software Development Engineer

Kickdrum

Jul 2021 - May 2022

(11 MONTHS)

- Delivered a **meeting-recording bot UI** in **React.js** → automated attendance; manual attendance time halved (**−50%**).
- Worked on **Chrome extension** features that summarized discussions → **document creation −40%** through auto-highlighting key points.
- Designed & tested an **Azure YAML pipeline** for a **.NET 3** project → **100% pass rate** across **50+ SonarQube** test scenarios.
- Raised CI/CD reliability with **85%+ test coverage**, ensuring consistent, successful builds.

SKILLS

Languages

JavaScript

TypeScript

Python

HTML5

CSS / SCSS

Frontend

React.JS

Next.JS

Tailwind CSS

Backend

Node.JS/Nest.JS

Flask

REST APIs

Supabase

Cloud & DevOps

GIT

AWS

Azure

CI/CD (YAML)

SonarQube

Analytics & Tools

Mixpanel

Figma

Chroma

BM25

AI/LLM

Generative AI

LangChain

RAG

PROJECT

TalentLens AI — LLM-Powered Candidate ATS (RAG + LangChain)

Sep 2025

Stack : RAG, LangChain, Groq (Llama 3.1), Chroma, BM25, Streamlit

[Live Demo](#)

- Built an **AI-powered ATS** that critiques resumes line-by-line, rewrites vague points, and suggests best-fit roles and companies using **RAG + Llama 3.1**.
- Implemented **hybrid retrieval (BM25 + embeddings)** to ground feedback in real resume and JD context, ensuring truthful and specific rewrites.
- Designed **structured LLM pipelines** in **LangChain** for JSON-based outputs, enabling consistent reasoning and line-level citations.
- Generated **JD-tailored resume redrafts** with factual placeholders and predicted **top-3 matching roles/companies** using semantic similarity.

CERTIFICATION

IBM – Generative AI Engineering Certification

Sep 2025

EDUCATION

Kiit University

2017 - 2021

BTech in Computer Science and Communication Engineering